# Six math clues · answer key

What holds a paper boat?

Age 8 / Grade 2 stretch. Offer tens-and-ones drawings, repeated addition, or oral reasoning.

### Math 1 · 45

8 + 7 = 15. Add 2 tens, 1 ten, and the new ten: 28 + 17 = 45.

### Math 2 · 26

52 is 4 tens and 12 ones. 12 − 6 = 6; 4 − 2 = 2. So 52 − 26 = 26.

### Math 3 · 63

6 + 7 = 13. 3 tens + 2 tens + 1 ten = 6 tens. So 36 + 27 = 63.

### Math 4 · 33

11 − 8 = 3; 6 − 3 = 3. So 71 − 38 = 33.

### Math 5 · 43

4 + 9 = 13. 2 tens + 1 ten + 1 ten = 4 tens. So 24 + 19 = 43.

### Math 6 · 35

13 − 8 = 5; 5 − 2 = 3. So 63 − 28 = 35.

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Age 8; Grade 2 math, number-crunch template. Six two-digit regrouping equations. Offer tens-and-ones sketches, counters, spoken reasoning, or a pause between pages.

# Reading & writing · answer key

What holds a paper boat?

### Reading · 1

Water pushed up on the boat. The boat pushed water aside, or displaced it. When the upward push balanced the boat's weight, it floated.

### Reading · 2

Pennies increased the load. The boat sank lower to displace more water, so water reached higher on its sides.

### Reading · 3

Plant fibers absorbed water. The paper became heavier and softer and lost its shape. A crayon or wax coat might slow soaking.

### Writing

Accept one sensible command, spoken, dictated, or written. Example: "First fold the paper in half, then fold the corners down." A clear two-step sequence is welcome.

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The child is age 8: Grade 3 reading and Grade 2 math. Welcome shared reading and dictation. Buoyancy is the upward force exerted by water; for a floating boat it balances total weight. The air-filled interior helps exclude water, rather than making air a source of upward force. A flat dry sheet can float briefly; paper type, folds, load, tipping and soaking affect the result. The story includes selected folding commands, not a complete boat-making guide. All math counts and diorama loads and times are invented comparisons, not real capacities. An at-home try is optional: use a tub or sink with an adult nearby.

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Accept the child's spoken command, dictation, invented spelling, or writing. One clear sentence is enough. Example: "First fold the paper in half, then fold the corners down."

# Four clues to keep

What holds a paper boat?

### Quiz 1. What holds a floating boat up?

Water pushes up. The push equals the weight of the displaced water. A floating boat displaces enough water for that push to balance its weight.

### Quiz 2. How do folded walls and a broad bottom help?

The walls keep water out. The broad bottom and air-filled space let the boat displace enough water to support its weight.

### Quiz 3. What happens to the waterline when Little adds pennies?

The load grows. The boat sinks lower and displaces more water, so the waterline climbs. Too much load can let water spill in.

### Quiz 4. Why does the paper boat soften over time?

Tiny plant fibers absorb water. Wet paper becomes heavier and softer and can lose its shape. A crayon or wax coat slows soaking.

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The child is age 8: Grade 3 reading and Grade 2 math. Welcome shared reading and dictation. Buoyancy is the upward force exerted by water; for a floating boat it balances total weight. The air-filled interior helps exclude water, rather than making air a source of upward force. A flat dry sheet can float briefly; paper type, folds, load, tipping and soaking affect the result. The story includes selected folding commands, not a complete boat-making guide. All math counts and diorama loads and times are invented comparisons, not real capacities. An at-home try is optional: use a tub or sink with an adult nearby.